Hengyu Lian

Chapel Hill, North Carolina

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Research Interests

My research lies at the intersection of physical optics and computer science, including:

- **Computational imaging:** computational cameras and photography (privacy-preserving vision, phase imaging, etc.)
- **Computational displays:** holographic and near-eye displays for AR/VR
- **Optical field manipulation:** wavefront shaping

Education

- **University of North Carolina at Chapel Hill** — **Chapel Hill, NC**
  Doctor of Philosophy — *August 2024 – May 2029 (expected)*
  Major: Computer Science
  Advisor: ***Prof. Praneeth Chakravarthula***
- **Shanghai Jiao Tong University** — **Shanghai, China**
  Master of Science and Engineering — *September 2021 – June 2024*
  Major: Electronic and Information Engineering
  Advisor: ***Prof. Yuan Qu and Prof. Jiamiao Yang***
- **Tianjin University** — **Tianjin, China**
  Bachelor of Engineering — *September 2017 – July 2021*
  Major: Electronic Engineering
  Advisor: ***Prof. Min Lai***

Publications

† Equal contribution

- **Hengyu Lian**, Debabrata Mandal, Anzhou Wen, et al., "Conditional Imaging for Privacy-Preserving Visual Sensing", **arXiv preprint** arXiv:2610.28835, 2026.
- Zihao Zou, Jiaming Liu, **Hengyu Lian**, et al., "Asynchronous Multimodal View Synthesis with Probabilistic Temporal Encoding", **SIGGRAPH Asia** Conference Papers, 2026.
- Huaizhi Qu†, Yujie Wang†, Ruichen Zhang, **Hengyu Lian**, et al., "HoloZip: High Hologram Compression via Latent-of-Latent Coding", **IEEE International Conference on Computational Photography (ICCP)**, 2025.
- Huazhen Liu, Rongjun Shao, Yuan Qu, Chunxu Ding, **Hengyu Lian**, et al., "High-accuracy phase unwrapping based on binarized wrap count", **Optics Express**, 32(25):44605–44623, 2024.
- Yuan Qu†, **Hengyu Lian†**, Rongjun Shao, et al., "Time series analysis for financial indices using optical reservoir computing", **Optical Engineering**, 63(5):054108, 2024.
- Yuan Qu†, **Hengyu Lian†**, Chunxu Ding, et al., "High-frame-rate reconfigurable diffractive neural network based on superpixels", **Optics Letters**, 48(19):5025–5028, 2023.

Skills

- **Optics & algorithms:** scalar diffraction modeling, phase retrieval, computer-generated holography
- **Hardware:** SLMs, DMDs, cameras and image sensors, optical setup and alignment, SolidWorks
- **Programming:** Python, MATLAB, PyTorch, Git, Linux

Work Experience and Services

- **Reviewer:**
  Optics Letters, Optics Express
- **Graduate Research Assistant:** *June 2026 – Present*
  Visual Computing and Augmented Intelligence Lab
  Advisor: Prof. Praneeth Chakravarthula
- **Teaching Assistant:** *August 2025 – May 2026*
  Course: Americans with Disabilities Act (ADA)
  Advisor: Prof. Brent Munsell
- **Teaching Assistant:** *August 2024 – May 2025*
  Course: COMP 590-059 Programming Methods, Models, Languages and Analysis
  Advisor: Prof. David Stotts
- **Graduate Research Assistant:** *September 2021 – June 2024*
  Intelligent Photoelectric Sensing Institute
  Advisor: Prof. Yuan Qu and Prof. Jiamiao Yang

Awards

- Guo Xie Birong Scholarship awarded by Shanghai Jiao Tong University, 10/2023 (Rank: 1/62)
- First-Class Scholarship awarded by Shanghai Jiao Tong University, 11/2023, 11/2022 (top 30%)